REVA UNIVERSITY
Bengaluru, India

REVA Academy for Corporate Excellence

SRN	AI09							

Name: Hari Krishnan

Program: MSc in Artificial Intelligence	Course: Deep Learning
Date: 23rd August 2025	Time: 10:00am to 1:00pm

Answers:
Section: A – UNIT I
Q2. a. Dataset: Use the IMDB Movie Reviews Dataset available in keras.datasets.imdb (50,000 reviews labeled as positive or negative). Load and preprocess the IMDB dataset. Show how reviews are represented as integer sequences. Demonstrate one-hot encoding of the top 10,000 words and comment on the sparsity of the resulting representation b. Using the same dataset, implement word embeddings (via Embedding layer in Keras or pretrained Word2Vec). Compare the dense embeddings with the one-hot encoded representation by visualizing similarities (e.g., most similar words to "good" or "bad"). c. Train a simple neural network classifier twice: once with one-hot encoded vectors, and once with embeddings. Report validation accuracy in both cases and critically evaluate why embeddings improve performance for text classification.
Ans: Added in **Module7_End_Exam_Deep_Learning_HariKrishnan.pynb** file
Section: B – UNIT II
Q3. a. Using the IMDB Movie Reviews Dataset (Keras), preprocess the text and apply Word2Vec or Doc2Vec embeddings to represent reviews numerically. Show how the embeddings capture semantic meaning. b. Train two sequence models: 1. A simple RNN 2. An LSTM on the IMDB dataset for sentiment classification. Compare their training/validation performance and highlight differences. c. Using Hugging Face's pretrained model bart-large-mnli, apply zero-shot text classification on 10 review sentences. Compare the predictions with your RNN/LSTM models and critically evaluate the trade-offs of using pretrained LLMs vs training models from scratch.
Ans: Added in **Module7_End_Exam_Deep_Learning_HariKrishnan.pynb** file

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Section: C – UNIT III
Q6. a. Dataset: Use the CIFAR-100 dataset available in keras.datasets.cifar100. (a) Build and train a basic CNN from scratch (3–4 convolutional + pooling layers) to classify CIFAR-100 images. Report training and validation accuracy. b. Apply data augmentation (rotation, flipping, cropping, etc.) and re-train the same CNN. Compare results with the non-augmented model. c. Critically discuss the challenges of training CNNs on a dataset with a large number of classes (100). Suggest at least two strategies (e.g., deeper architectures, transfer learning, label hierarchy) to address these issues.
Ans: Added in **Module7_End_Exam_Deep_Learning_HariKrishnan.pynb** file
Section: D – UNIT IV
Q8. a. Dataset Description: A collection of ~1,500 wildlife images featuring four animal classes — buffalo, elephant, rhino, and zebra. https://www.kaggle.com/datasets/ankanghosh651/object-detection-wildlife-dataset-yolo-format Load the Wildlife dataset and train a YOLOv8 model to detect the four animal classes. Display detection results on at least 5 test images. b. Track key metrics during training — specifically mAP@0.5 and mAP@0.5:0.95 — and interpret what they indicate about the model's detection performance. c. Propose at least two enhancements (e.g., data augmentation strategies, tuning architecture hyperparameters, or fine-tuning pre-trained weights) that could significantly improve detection accuracy. Justify your choices with reasoning or code examples.
Ans: Added in **Module7_End_Exam_Deep_Learning_HariKrishnan.pynb** file

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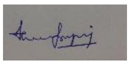
STUDENT UNDERTAKING

I, **[Hari Krishnan]**, bearing **Student ID/Registration Number: [AI09]**, hereby declare that:

- All answers provided in this answer script are **original** and a result of my own understanding.
- I have not engaged in any form of **copying, plagiarism, impersonation, or malpractice**.
- I understand that **any academic dishonesty** will lead to disciplinary action as per university regulations.



Signature: ____________________

Date: ____23-Aug-2025 ________

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